Mathematical Analysis Exam

Year 1 Computer Science

June 7, 2021

Let ✤ be the first letter of you FAMILY NAME,

i.e., for Mircea Ivan, ✤$=I$

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SUBJECT 1, YEAR 1C

1 **Calculate (using integrals) the volume of the tetrahedron of vertices** $\text{✽}_1(0,0,0)$, $\text{✽}_2(a,0,0)$, $\text{✽}_3(0,b,0)$, $\text{✽}_4(0,0,c)$, $\quad a,b,c>0$.

2 **Imagine a Chebyshev type inequality for improper integrals.**

3 $$\int_0^\infty \frac{\frac{1}{1+x^2}-\frac{1}{1+x^3}}{x}\,\mathrm{d}x = ? \qquad \int_0^\infty \frac{\frac{1}{1+x^{\text{✽}}}-e^{-x}}{x}\,\mathrm{d}x = ?$$

SUBJECT 2, YEAR 1C

1 **Find the area of the region defined by:**

$1 \leq xy \leq \ast$ **and** $x \leq y \leq \ast\, x, \quad \ast > 0.$

...

2 $x^2 y'(x) - 2xy(x) = y(x)^2, \qquad y(0) = 1.$

3 $$\int_0^{\infty} \frac{-e^{-x} - \log(x+1) + 1}{x^2}\, \mathrm{d}x = ?$$

SUBJECT 3, YEAR 1C

1 $$\int_0^\infty \frac{e^{-ax} - \frac{1}{(x+1)^a}}{x}\,\mathrm{d}x = ?;$$

$$\lim_{a\to\infty}\left(\psi(a) - \log(a)\right) = ?$$

2 $$\int_0^1 \frac{\log x}{\sqrt{1-x}}\,\mathrm{d}x = ?$$